Chapter 5. Entire Functions

Jensen's formula, order of growth, infinite products, Weierstrass and Hadamard factorization

Contents

Overview

Entire functions are holomorphic in the whole complex plane. The chapter is organized around three questions.

1. *Where can they vanish?* The obvious necessary condition is sufficient: for any sequence $\{z_n\}$ with no limit point in $\mathbb{C}$ there is an entire function vanishing exactly there. The construction is inspired by Euler's product for $\sin \pi z$, refined by the *Weierstrass canonical factors*.
2. *How do they grow at infinity?* The larger a function is, the more zeros it can have; for a polynomial of degree d, $\sup_{|z|=R}|P(z)| \approx R^d$ and P has d zeros. A precise version is *Jensen's formula*, linking the number of zeros in a disc to the logarithmic average of $|f|$ on the circle.
3. *To what extent are they determined by their zeros?* If f has finite order, it is determined by its zeros up to a simple factor: the *Hadamard factorization theorem*.

1 Jensen's formula

Motivation. How many zeros can an entire function have in a disc of radius R? Jensen's formula relates the zeros of f to the average of $\log|f|$ on the boundary circle, so growth controls the number of zeros.

D_R and C_R denote the open disc and circle of radius R centered at 0. We exclude the function that vanishes identically.

Theorem 1.1 (Jensen's formula)

Let Ω be an open set that contains the closure of a disc D_R and suppose that f is holomorphic in Ω, $f(0) \neq 0$, and f vanishes nowhere on the circle C_R. If $z_1, \dots, z_N$ denote the zeros of f inside the disc (counted with multiplicities), then

$$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta. \tag{1}$$

Proof.

Step 1. If f_1, f_2 satisfy the hypotheses and (1), so does $f_1 f_2$, since $\log xy = \log x + \log y$ for $x, y > 0$ and the zeros of $f_1 f_2$ are the union of those of f_1 and f_2.

Step 2. $g(z) = f(z)/((z - z_1)\cdots(z - z_N))$, defined on $\Omega - \{z_1, \dots, z_N\}$, is bounded near each z_j, so each z_j is a removable singularity: $f(z) = (z - z_1)\cdots(z - z_N)g(z)$ with g holomorphic in Ω and nowhere vanishing on $\overline{D_R}$. By Step 1 it suffices to prove (1) for such g and for functions $z - z_j$.

Step 3. For g nowhere vanishing on $\overline{D_R}$ we must show $\log|g(0)| = \frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta$. In a slightly larger disc, $g = e^h$ with h holomorphic (discs are simply connected; Theorem 6.2). Then $|g| = e^{\mathrm{Re}(h)}$, so $\log|g| = \mathrm{Re}(h)$, and the mean value property (Corollary 7.3) gives the formula.

Step 4. For $f(z) = z - w$, $w \in D_R$, we must show $\log|w| = \log\frac{|w|}{R} + \frac{1}{2\pi}\int_0^{2\pi} \log|Re^{i\theta} - w|\, d\theta$. Since $\log(|w|/R) = \log|w| - \log R$ and $\log|Re^{i\theta} - w| = \log R + \log|e^{i\theta} - w/R|$, it suffices to show $\int_0^{2\pi} \log|e^{i\theta} - a|\, d\theta = 0$ for $|a| < 1$, equivalently (via $\theta \mapsto -\theta$, $|e^{i\theta} - a| = |1 - ae^{-i\theta}|$) $\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0$. $F(z) = 1 - az$ vanishes nowhere on the closed unit disc, so $F = e^G$ in a disc of radius > 1 and $\log|F| = \mathrm{Re}(G)$ is harmonic. Since $\log|F(0)| = 0$, the mean value property gives the result. ∎

For f holomorphic on $\overline{D_R}$, let $n(r) = n_f(r)$ be the number of zeros of f (with multiplicities) in D_r,

$0 < r < R$; $n(r)$ is non-decreasing. If $f(0) \neq 0$ and f does not vanish on C_R, then

$$\int_0^R n(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\,d\theta - \log|f(0)|, \tag{2}$$

which follows from Jensen's formula and:

Lemma 1.2

If $z_1, \ldots, z_N$ are the zeros of f inside the disc D_R, then

$$\int_0^R n(r)\frac{dr}{r} = \sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right|.$$

Proof. $\sum_{k=1}^N \log\left|\frac{R}{z_k}\right| = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r}$. With $\eta_k(r) = 1$ if $r > |z_k|$ and 0 otherwise, $\sum_k \eta_k(r) = n(r)$, so $\sum_k \int_{|z_k|}^R \frac{dr}{r} = \sum_k \int_0^R \eta_k(r)\frac{dr}{r} = \int_0^R n(r)\frac{dr}{r}$. ∎

2 Functions of finite order

Motivation. Measuring growth by an order ρ with $|f(z)| \leq Ae^{B|z|^\rho}$, Jensen's formula gives a quantitative bound on the number of zeros. This prepares the factorization theorems of the chapter.

Definition (order of growth)

An entire function f has an **order of growth** $\leq \rho$ if $|f(z)| \leq Ae^{B|z|^\rho}$ for all $z \in \mathbb{C}$, for some constants $A, B > 0$. The **order of growth** of f is $\rho_f = \inf \rho$ over all such $\rho > 0$.

For example, e^{z^2} has order of growth 2.

Theorem 2.1

If f is an entire function that has an order of growth $\leq \rho$, then:
(i) $n(r) \leq Cr^\rho$ for some $C > 0$ and all sufficiently large r.
(ii) If $z_1, z_2, \ldots$ denote the zeros of f, with $z_k \neq 0$, then for all $s > \rho$ we have $\displaystyle\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty$.

Proof. It suffices to treat $f(0) \neq 0$: otherwise use $F(z) = f(z)/z^\ell$, ℓ the order of the zero at 0; n_f and n_F differ by a constant and F also has order of growth $\leq \rho$. By (2) with $R = 2r$,

$$\int_r^{2r} n(x)\frac{dx}{x} \leq \int_0^{2r} n(x)\frac{dx}{x} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(2re^{i\theta})|\,d\theta - \log|f(0)|.$$

Since n is increasing, the left side is $\geq n(r)\int_r^{2r}\frac{dx}{x} = n(r)\log 2$; and by the growth condition $\int_0^{2\pi}\log|f(2re^{i\theta})|\,d\theta \leq \int_0^{2\pi}\log|Ae^{B(2r)^\rho}|\,d\theta \leq C'r^\rho$ for large r. Hence $n(r) \leq Cr^\rho$. For (ii),

$$\sum_{|z_k|\geq 1} |z_k|^{-s} = \sum_{j=0}^{\infty} \sum_{2^j \leq |z_k| < 2^{j+1}} |z_k|^{-s} \leq \sum_{j=0}^{\infty} 2^{-js} n(2^{j+1}) \leq c\sum_{j=0}^{\infty} 2^{-js}2^{(j+1)\rho} \leq c'\sum_{j=0}^{\infty}(2^{\rho-s})^j < \infty,$$

since $s > \rho$. ∎

The condition $s > \rho$ cannot be improved:

Example 1

$f(z) = \sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i}$ satisfies $|f(z)| \le e^{\pi|z|}$, and taking $z = ix$ shows the order of growth is exactly 1. f vanishes to order 1 at each $n \in \mathbb{Z}$, and $\sum_{n\neq 0} 1/|n|^s < \infty$ precisely when $s > 1$.

Example 2

$f(z) = \cos z^{1/2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{(2n)!}$ is entire with $|f(z)| \le e^{|z|^{1/2}}$, of order $1/2$. It vanishes at $z_n = ((n + 1/2)\pi)^2$, and $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

Given $z_1, z_2, \ldots$, is there an entire f with zeros exactly there? Necessarily $\lim_{k\to\infty} |z_k| = \infty$ (Theorem 4.8). Weierstrass showed this suffices, by inserting factors into $(z-z_1)(z-z_2)\cdots$ that guarantee convergence without introducing new zeros.

3 Infinite products

Motivation. A polynomial is determined by its zeros up to a constant, through $\prod(z - a_k)$. To do the same for entire functions with infinitely many zeros we need infinite products and conditions for their convergence.

3.1 Generalities

For complex a_n, the product $\prod_{n=1}^{\infty}(1 + a_n)$ **converges** if $\lim_{N\to\infty} \prod_{n=1}^{N}(1 + a_n)$ exists.

Proposition 3.1

If $\sum |a_n| < \infty$, then the product $\prod_{n=1}^{\infty}(1 + a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.

Proof. For large n, $|a_n| < 1/2$; discarding finitely many terms, assume this for all n. Define $\log(1 + a_n)$ by the power series (6) of Chapter 3, so $1 + z = e^{\log(1+z)}$ for $|z| < 1$. Then $\prod_{n=1}^{N}(1 + a_n) = \prod_{n=1}^{N} e^{\log(1+a_n)} = e^{B_N}$ with $B_N = \sum_{n=1}^{N} b_n$, $b_n = \log(1 + a_n)$. From the power series, $|\log(1 + z)| \le 2|z|$ for $|z| < 1/2$, so $|b_n| \le 2|a_n|$ and $B_N \to B$. By continuity of the exponential, $e^{B_N} \to e^B$. If $1 + a_n \neq 0$ for all n, the limit is $e^B \neq 0$. ∎

Proposition 3.2

Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω. If there exist constants $c_n > 0$ such that $\sum c_n < \infty$ and $|F_n(z) - 1| \le c_n$ for all $z \in \Omega$, then:
(i) The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$.
(ii) If $F_n(z)$ does not vanish for any n, then

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F_n'(z)}{F_n(z)}.$$

Proof. (i) Write $F_n(z) = 1 + a_n(z)$ with $|a_n(z)| \le c_n$ and argue as in Proposition 3.1; the estimates are uniform in z since the c_n are constants. The uniform limit is holomorphic (Theorem 5.2).

(ii) Let $K \subset \Omega$ be compact and $G_N = \prod_{n=1}^{N} F_n$. Since $G_N \to F$ uniformly, Theorem 5.3 gives $G_N' \to F'$ uniformly on K. G_N is uniformly bounded below on K, so $G_N'/G_N \to F'/F$ uniformly on K; and $G_N'/G_N = \sum_{n=1}^{N} F_n'/F_n$ (Section 4). ∎

3.2 Example: the product formula for the sine function

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \Big(1 - \frac{z^2}{n^2}\Big) \tag{3}$$

for all $z \in \mathbb{C}$; it is derived from the sum formula for the cotangent

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\le N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}, \tag{4}$$

valid for $z \notin \mathbb{Z}$. The halves of $\sum 1/(z+n)$ with $n > 0$ and $n < 0$ diverge separately; only the symmetric interpretation converges.

Proof of (4). $F(z) = \pi \cot \pi z$ has the properties:

(i) $F(z+1) = F(z)$ for $z \notin \mathbb{Z}$;

(ii) $F(z) = \frac{1}{z} + F_0(z)$ with F_0 analytic near 0;

(iii) F has simple poles at the integers and no other singularities. The function $\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ has the same three properties: for (i),

$$\sum_{|n|\le N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n|\le N} \frac{1}{z+n},$$

and let $N \to \infty$; (ii), (iii) are evident from $\frac{1}{z} + \sum_{n\ge 1} \frac{2z}{z^2-n^2}$. Hence $\Delta(z) = F(z) - \sum \frac{1}{z+n}$ is periodic with period 1, has a removable singularity at 0 by (ii), hence at all integers by periodicity, and is entire. It suffices to show Δ is bounded, hence (by periodicity) bounded in the strip $|\mathrm{Re}(z)| \le 1/2$. It is bounded in the rectangle $|\mathrm{Im}(z)| \le 1$. For $z = x + iy$ with $y > 1$,

$$\pi \cot \pi z = \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \pi i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}}$$

is bounded, and

$$\Big|\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}\Big| = \Big|\frac{1}{x+iy} + \sum_{n=1}^{\infty} \frac{2(x+iy)}{x^2 - y^2 - n^2 + 2ixy}\Big| \le C + C\sum_{n=1}^{\infty} \frac{y}{y^2+n^2}.$$

The last sum is majorized by $\int_0^{\infty} \frac{y}{y^2+x^2}\,dx$ (the integrand is decreasing in x), which is independent of y (change variables $x \mapsto yx$). Similarly for $y < -1$. So Δ is bounded on $\mathbb{C}$, hence constant by Liouville; Δ is odd, so the constant is 0. ∎

Proof of (3). Let $G(z) = \sin \pi z$ and $P(z) = \pi z \prod_{n=1}^{\infty}(1 - z^2/n^2)$. Since $\sum 1/n^2 < \infty$, Proposition 3.2 shows P converges and, away from the integers, $\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}$. Since $G'/G = \pi \cot \pi z$, (4) gives

$$\Big(\frac{P}{G}\Big)' = \frac{P}{G}\Big[\frac{P'}{P} - \frac{G'}{G}\Big] = 0,$$

so $P = cG$. Dividing by z and letting $z \to 0$, $c = 1$. ∎

4 Weierstrass infinite products

Motivation. The naive product $\prod(1 - z/a_n)$ may diverge. Weierstrass's canonical factors repair this, so that an entire function with any prescribed sequence of zeros (tending to infinity) exists.

Theorem 4.1 (Weierstrass)

Given any sequence $\{a_n\}$ of complex numbers with $|a_n| \to \infty$ as $n \to \infty$, there exists an entire function f that vanishes at all $z = a_n$ and nowhere else. Any other such entire function is of the form $f(z)e^{g(z)}$, where g is entire.

Repetitions are allowed in $\{a_n\}$, so this gives entire functions with prescribed zeros *and multiplicities* (the multiplicity of a zero a is the m with $f(z) = (z-a)^m g(z)$, $g(a) \neq 0$).

Proof. If f_1, f_2 both vanish exactly at the a_n (with multiplicities), f_1/f_2 has removable singularities there, so it is entire and nowhere vanishing; hence $f_1/f_2 = e^g$ for an entire g (Theorem 6.2). This proves the last statement.

To construct f: the naive product $\prod(1 - z/a_n)$ converges only for suitable $\{a_n\}$, so we insert exponential factors. Define the **canonical factors** of degree k:

$$E_0(z) = 1 - z, \qquad E_k(z) = (1-z)e^{z+z^2/2+\cdots+z^k/k} \quad (k \geq 1).$$

Suppose f is to have a zero of order m at the origin and $a_1, a_2, \ldots$ are non-zero. The **Weierstrass product** is

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Fix $R > 0$ and $|z| < R$. Factors with $|a_n| \leq 2R$ are finitely many, and their product vanishes exactly at the a_n with $|a_n| < R$. If $|a_n| > 2R$ then $|z/a_n| \leq 1/2$, and by Lemma 4.2 (with c independent of n)

$$|1 - E_n(z/a_n)| \leq c\Big|\frac{z}{a_n}\Big|^{n+1} \leq \frac{c}{2^{n+1}}.$$

By Section 3, $\prod_{|a_n|>2R} E_n(z/a_n)$ is holomorphic and non-vanishing in $|z| < R$. As R is arbitrary, f has the required properties. ∎

Lemma 4.2

If $|z| \leq 1/2$, then $|1 - E_k(z)| \leq c|z|^{k+1}$ for some $c > 0$.

Proof. For $|z| \leq 1/2$, $1 - z = e^{\log(1-z)}$ (power series), so $E_k(z) = e^{\log(1-z)+z+z^2/2+\cdots+z^k/k} = e^w$ with $w = -\sum_{n=k+1}^{\infty} z^n/n$. Since $|z| \leq 1/2$,

$$|w| \leq |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \leq |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \leq 2|z|^{k+1}.$$

In particular $|w| \leq 1$, so $|1 - E_k(z)| = |1 - e^w| \leq c'|w| \leq c|z|^{k+1}$. ∎

Remark

The constant c is independent of k: one may take $c' = e$, $c = 2e$.

5 Hadamard's factorization theorem

Weierstrass's theorem writes a function with zeros $a_1, a_2, \ldots$ as $e^{g(z)} z^m \prod E_n(z/a_n)$. Hadamard showed that for functions of finite order the degree of the canonical factors can be taken constant, and g is then a polynomial. Recall that if f has order of growth $\leq \rho$, then $n(r) \leq Cr^\rho$ for large r and $\sum |a_n|^{-s} < \infty$ for $s > \rho$ (Theorem 2.1).

Theorem 5.1 (Hadamard)

Suppose f is entire and has growth order ρ_0. Let k be the integer so that $k \le \rho_0 < k+1$. If $a_1, a_2, \dots$ denote the (non-zero) zeros of f, then

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\le k$, and m is the order of the zero of f at $z = 0$.

Main lemmas.

Lemma 5.2

The canonical products satisfy

$$|E_k(z)| \ge e^{-c|z|^{k+1}} \text{ if } |z| \le 1/2 \qquad \text{and} \qquad |E_k(z)| \ge |1-z| e^{-c'|z|^k} \text{ if } |z| \ge 1/2.$$

Proof. If $|z| \le 1/2$, $E_k(z) = e^{\log(1-z) + \sum_{n=1}^{k} z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w$, and $|e^w| \ge e^{-|w|}$ with $|w| \le c|z|^{k+1}$. If $|z| \ge 1/2$, $|E_k(z)| = |1-z||e^{z+z^2/2+\cdots+z^k/k}|$, and $|e^{z+\cdots+z^k/k}| \ge e^{-|z+\cdots+z^k/k|} \ge e^{-c'|z|^k}$ (as $|z| \ge 1/2$). ■

The key is a lower bound for the product of canonical factors away from the zeros.

Lemma 5.3

For any s with $\rho_0 < s < k+1$, we have

$$\Big| \prod_{n=1}^{\infty} E_k(z/a_n) \Big| \ge e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \dots$.

Proof. Write $\prod_n E_k(z/a_n) = \prod_{|a_n| \le 2|z|} E_k(z/a_n) \prod_{|a_n| > 2|z|} E_k(z/a_n)$. For the second product, with no restriction on z, Lemma 5.2 gives

$$\Big| \prod_{|a_n|>2|z|} E_k(z/a_n) \Big| \ge \prod_{|a_n|>2|z|} e^{-c|z/a_n|^{k+1}} = e^{-c|z|^{k+1} \sum_{|a_n|>2|z|} |a_n|^{-k-1}},$$

and since $|a_n| > 2|z|$ and $s < k+1$, $|a_n|^{-k-1} = |a_n|^{-s}|a_n|^{s-k-1} \le C|a_n|^{-s}|z|^{s-k-1}$; as $\sum |a_n|^{-s} < \infty$, this product is $\ge e^{-c|z|^s}$. For the first product, by the second part of Lemma 5.2,

$$\Big| \prod_{|a_n| \le 2|z|} E_k(z/a_n) \Big| \ge \prod_{|a_n| \le 2|z|} \Big|1 - \frac{z}{a_n}\Big| \prod_{|a_n| \le 2|z|} e^{-c'|z/a_n|^k}. \tag{5}$$

Here $\prod_{|a_n| \le 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k \sum_{|a_n| \le 2|z|} |a_n|^{-k}}$ and $|a_n|^{-k} = |a_n|^{-s}|a_n|^{s-k} \le C|a_n|^{-s}|z|^{s-k}$, so this is $\ge e^{-c|z|^s}$. For the first factor of (5) we need the restriction on z: if z is not in the disc of radius $|a_n|^{-k-1}$ about a_n, then $|a_n - z| \ge |a_n|^{-k-1}$, so

$$\prod_{|a_n| \le 2|z|} \Big|1 - \frac{z}{a_n}\Big| = \prod_{|a_n| \le 2|z|} \Big|\frac{a_n - z}{a_n}\Big| \ge \prod_{|a_n| \le 2|z|} |a_n|^{-k-1}|a_n|^{-1} = \prod_{|a_n| \le 2|z|} |a_n|^{-k-2}.$$

Finally $\log \prod_{|a_n| \le 2|z|} |a_n|^{k+2} = (k+2) \sum_{|a_n| \le 2|z|} \log|a_n| \le (k+2) n(2|z|) \log 2|z| \le c|z|^s \log 2|z| \le c'|z|^{s'}$ for any $s' > s$, using $n(2|z|) \le c|z|^s$ (Theorem 2.1). Since $s > \rho_0$ was arbitrary, starting with s close enough to ρ_0 gives the lemma (with s' in place of s). ■

Corollary 5.4

There exists a sequence of radii $r_1, r_2, \dots$ with $r_m \to \infty$, such that

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \geq e^{-c|z|^s} \qquad \text{for } |z| = r_m.$$

Proof. Since $\sum |a_n|^{-k-1} < \infty$, there is N with $\sum_{n\geq N} |a_n|^{-k-1} < 1/10$. Given consecutive large integers L, $L+1$, there is $r \in [L, L+1]$ such that the circle of radius r about 0 meets none of the forbidden discs of Lemma 5.3: otherwise the intervals $I_n = [|a_n| - |a_n|^{-k-1}, |a_n| + |a_n|^{-k-1}]$, of length $2|a_n|^{-k-1}$, would cover $[L, L+1]$, giving $2\sum_{n\geq N} |a_n|^{-k-1} \geq 1$, a contradiction. Apply Lemma 5.3 with $|z| = r$. ∎

Proof of Hadamard's theorem. Let $E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n)$. E is entire, as in the proof of Theorem 4.1: by Lemma 4.2, $|1 - E_k(z/a_n)| \leq c|z/a_n|^{k+1}$ for large n, and $\sum |a_n|^{-k-1}$ converges ($\rho_0 < s < k+1$). E has the zeros of f, so f/E is entire and nowhere vanishing: $f(z)/E(z) = e^{g(z)}$ for an entire g. Since f has growth order ρ_0 and by Corollary 5.4,

$$e^{\operatorname{Re}(g(z))} = \Big|\frac{f(z)}{E(z)}\Big| \leq c' e^{c|z|^s} \qquad \text{for } |z| = r_m,$$

so $\operatorname{Re}(g(z)) \leq C|z|^s$ for $|z| = r_m$. Lemma 5.5 gives that g is a polynomial of degree $\leq s$, hence $\leq k$ (as $s < k+1$). ∎

Lemma 5.5

Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies $u(z) \leq Cr^s$ whenever $|z| = r$, for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\leq s$.

Proof. Write $g(z) = \sum_{n=0}^{\infty} a_n z^n$. By Theorem 7.1,

$$\frac{1}{2\pi}\int_0^{2\pi} g(re^{i\theta}) e^{-in\theta}\, d\theta = \begin{cases} a_n r^n & n \geq 0, \\ 0 & n < 0. \end{cases} \tag{6}$$

Taking complex conjugates, $\frac{1}{2\pi}\int_0^{2\pi} \overline{g(re^{i\theta})} e^{-in\theta}\, d\theta = 0$ for $n > 0$ (7). Since $2u = g + \overline{g}$, adding (6) and (7),

$$a_n r^n = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta}) e^{-in\theta}\, d\theta \quad (n > 0), \qquad 2\operatorname{Re}(a_0) = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})\, d\theta.$$

[Supplement]Since $\int_0^{2\pi} e^{-in\theta} d\theta = 0$ for $n > 0$, $a_n r^n = -\frac{1}{\pi}\int_0^{2\pi} [Cr^s - u(re^{i\theta})] e^{-in\theta}\, d\theta$, and $Cr^s - u \geq 0$ on the circles in question, so

$$|a_n| r^n \leq \frac{1}{\pi}\int_0^{2\pi} [Cr^s - u(re^{i\theta})]\, d\theta = 2Cr^s - 2\operatorname{Re}(a_0).$$

Letting $r \to \infty$ along the sequence, $a_n = 0$ for $n > s$. ∎

6 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 5 (book numbers kept): Wallis's product from the sine product, the order of the Jacobi theta function, and two applications of Hadamard's theorem.

Exercise 5.6 (Wallis's product)

(a) Prove Wallis's product formula

$$\frac{\pi}{2} = \prod_{m=1}^{\infty} \frac{2m \cdot 2m}{(2m-1)(2m+1)} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdots .$$

(b) Deduce that $\lim_{n\to\infty} \dfrac{2^{2n}(n!)^2}{(2n)!\,(2n+1)^{1/2}} = \sqrt{\pi/2}$.

Solution.

(a) Put $z = 1/2$ in the product formula (3) for the sine:

$$1 = \sin\frac{\pi}{2} = \frac{\pi}{2} \prod_{m=1}^{\infty} \Big(1 - \frac{1}{4m^2}\Big).$$

So the partial products $P_N = \prod_{m\le N}(1 - \frac{1}{4m^2})$ converge to $2/\pi \neq 0$. Hence $1/P_N \to \pi/2$, and

$$\frac{1}{P_N} = \prod_{m=1}^{N} \frac{4m^2}{4m^2-1} = \prod_{m=1}^{N} \frac{2m \cdot 2m}{(2m-1)(2m+1)}.$$

(b) Let $W_N = 1/P_N$.

Step 1. We have $\prod_{m\le N} 2m = 2^N N!$. The odd numbers below $2N$ and below $2N+2$ have products

$$\prod_{m\le N} (2m-1) = \frac{(2N)!}{2^N N!}, \qquad \prod_{m\le N} (2m+1) = \frac{(2N+1)!}{2^N N!}.$$

Step 2. Therefore

$$W_N = \frac{(2^N N!)^4}{(2N)!\,(2N+1)!} = \frac{2^{4N}(N!)^4}{((2N)!)^2(2N+1)} = \Big(\frac{2^{2N}(N!)^2}{(2N)!\,(2N+1)^{1/2}}\Big)^2.$$

Step 3. The bracket is positive and its square tends to $\pi/2$ by (a), so it tends to $\sqrt{\pi/2}$. ∎

Exercise 5.3 (Order of the theta function)

Fix τ with $t = \operatorname{Im}\tau > 0$. Show that the Jacobi theta function

$$\Theta(z \mid \tau) = \sum_{n=-\infty}^{\infty} e^{i\pi n^2 \tau} e^{2\pi i n z}$$

is an entire function of z of order exactly 2.

Solution.

Step 1. (Upper bound.) The n-th term has modulus $e^{-\pi n^2 t - 2\pi n \operatorname{Im} z} \le e^{-\pi n^2 t + 2\pi |n||z|}$. If $|n| \ge 4|z|/t$, then $2\pi|n||z| \le \pi n^2 t/2$, and the term is $\le e^{-\pi n^2 t/2}$. If $|n| < 4|z|/t$, the term is $\le e^{2\pi|n||z|} \le e^{8\pi|z|^2/t}$, and there are at most $8|z|/t + 1$ such n. Hence

$$|\Theta(z \mid \tau)| \le \Big(\frac{8|z|}{t} + 1\Big) e^{8\pi|z|^2/t} + \sum_n e^{-\pi n^2 t/2} \le A e^{B|z|^2}.$$

The series converges uniformly on compact sets by the same bound, so Θ is entire (Theorem 5.2) of order ≤ 2.

Step 2. (Quasi-periodicity.) Replacing n by $n+1$ in the sum,

$$\Theta(z+\tau \mid \tau) = \sum_n e^{i\pi(n^2+2n)\tau} e^{2\pi i n z} = e^{-i\pi\tau} e^{-2\pi i z} \sum_n e^{i\pi(n+1)^2\tau} e^{2\pi i (n+1) z} = e^{-i\pi\tau - 2\pi i z}\, \Theta(z \mid \tau).$$

Step 3. ($\Theta \not\equiv 0$.) On the real line the series converges uniformly, so $\int_0^1 \Theta(x \mid \tau)\, dx = 1$ (only $n = 0$ survives). Pick z_0 with $\Theta(z_0 \mid \tau) \neq 0$ and write $y_0 = \operatorname{Im} z_0$.

Step 4. (Lower bound.) Let $z_m = z_0 + m\tau$. Since $|e^{-i\pi\tau - 2\pi i z}| = e^{\pi t + 2\pi \operatorname{Im} z}$ and $\operatorname{Im} z_j = y_0 + jt$, Step 2 gives

$$|\Theta(z_m \mid \tau)| = |\Theta(z_0 \mid \tau)| \prod_{j=0}^{m-1} e^{\pi t + 2\pi(y_0 + jt)} = |\Theta(z_0 \mid \tau)|\, e^{\pi t m^2 + 2\pi y_0 m}.$$

If $|\Theta| \le A e^{B|z|^\rho}$ with $\rho < 2$, then, since $|z_m| \le |z_0| + m|\tau|$,

$$\pi t m^2 + 2\pi y_0 m + \log|\Theta(z_0 \mid \tau)| \le \log A + B(|z_0| + m|\tau|)^\rho,$$

which fails for large m. Hence the order is exactly 2. ∎

Exercise 5.11 (Picard's little theorem for finite order)

Show that an entire function of finite order that omits two distinct values is constant.

Solution.

Step 1. Let f have order $\le \rho$ and omit $a \neq b$. Then $g = f - a$ is entire and nowhere zero, and $|g| \le |f| + |a| \le (A + |a|)e^{B|z|^\rho}$, so g has finite order. By Hadamard's theorem (Theorem 5.1) with no zeros, $g = e^p$ with p a polynomial.

Step 2. Since f omits b, $e^{p(z)} \neq b - a$ for all z. As $b - a \neq 0$, there is $w \in \mathbb{C}$ with $e^w = b - a$.

Step 3. If p were non-constant, the fundamental theorem of algebra (Corollary 4.6) applied to $p - w$ would give z with $p(z) = w$, hence $e^{p(z)} = b - a$, a contradiction. So p is constant, and so is $f = a + e^p$. ∎

Exercise 5.13 (Solutions of $e^z = z$)

Show that the equation $e^z - z = 0$ has infinitely many solutions in $\mathbb{C}$.

Solution.

Step 1. $f(z) = e^z - z$ satisfies $|f| \le e^{|z|} + |z| \le 2e^{|z|}$, so its order is $\rho_0 \le 1$, and the integer k of Theorem 5.1 is 0 or 1. Also $f(0) = 1 \neq 0$.

Step 2. Suppose f had only finitely many zeros $a_1, \dots, a_N$. By Hadamard's theorem, $f = e^P \prod_{n \le N} E_k(z/a_n)$ with $\deg P \le k \le 1$. Each $E_k(z/a_n)$ is $(1 - z/a_n)$ times the exponential of a polynomial of degree $\le k$. Collecting factors,

$$e^z - z = Q(z) e^{\alpha z + \beta}$$

for a non-zero polynomial Q and constants $\alpha, \beta \in \mathbb{C}$.

Step 3. (Behaviour as $x \to -\infty$.) For real x, $|Q(x)e^{\alpha x + \beta}| = |Q(x)| e^{x \operatorname{Re}\alpha + \operatorname{Re}\beta}$, while $|e^x - x| = |x| + o(1)$ as $x \to -\infty$. If $\operatorname{Re}\alpha > 0$ the right side tends to 0; if $\operatorname{Re}\alpha < 0$ it grows faster than every power of $|x|$. Both contradict $|e^x - x| \sim |x|$. Hence $\operatorname{Re}\alpha = 0$.

Step 4. (Behaviour as $x \to +\infty$.) Then $|Q(x)e^{\alpha x + \beta}| = |Q(x)| e^{\operatorname{Re}\beta}$ grows at most polynomially. But $e^x - x \ge e^x/2$ for large x, a contradiction. So f has infinitely many zeros. ∎